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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

EC3003-1 - DIGITAL SIGNAL PROCESSING
 (For EC / AC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- If the DFT of a sequence $x(n)$ is $X(k)$, then the DFT of $x((n-2))_4$ is

A) $X(k)W_4^{2k}$	B) $x(k)W_4^{2k}$
C) $X(l)W_4^{4k}$	D) $X(k)W_4^{-2k}$
- An even function is said to be

A) Antisymmetric	B) Non-periodic
C) Periodic	D) Symmetric
- Frequency selectivity characteristics of DFT refers to

A) Ability to resolve different frequency components from input signal	B) Ability to translate into frequency domain
C) Ability to convert into discrete signal	D) None
- Parseval's theorem is defined as

A) $\sum_{n=0}^{N-1} x(n) ^2 = \sum_{k=0}^{N-1} X(k) ^2$	B) $\frac{1}{N} \sum_{n=0}^{N-1} x(n) ^2 = \sum_{k=0}^{N-1} X(k) ^2$
C) $\sum_{n=0}^{N-1} x(n) ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X(k) ^2$	D) $\sum_{n=0}^{N-1} x(n) ^2 = \frac{1}{N^2} \sum_{k=0}^{N-1} X(k) ^2$
- FFT algorithm was invented by

A) John Widmer	B) Cooley-Tukey
C) Joseph Fourier	D) Joseph Swan
- For a 16-point DFT number of complex multiplications and complex additions required are

A) 32 complex multiplications and 64 complex additions	B) 64 complex multiplications and 64 complex additions
C) 64 complex multiplications and 32 complex additions	D) 12 complex multiplications and 24 complex additions
- DITFFT algorithm divides the input sequence into

A) Positive and negative values	B) Smaller and larger numbers
C) Real and complex numbers	D) Even and odd samples
- FFT is a modification of DFT in terms of

A) Noise	B) Reduction in sample size
C) Computational speed	D) Reduction in stages

9. The overlap save method is used to calculate
- A) The discrete convolution between a sampled signal and a finite impulse response (FIR) filter
- B) The discrete convolution between a sampled signal and an infinite impulse response (IIR) filter
- C) The discrete convolution between a very long sequence and an infinite impulse response (IIR) filter
- D) The discrete convolution between a very long sequence and a finite impulse response (FIR) filter
10. In Chebyshev filter design approximation, the poles are plotted on the circumference of
- A) Ellipse
- B) Sphere
- C) Circle
- D) Hexagon
11. Identify the type of the filter that has the transfer function $|H(\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}}$
- A) Chebyshev Filters
- B) Butterworth Filters
- C) Elliptic Filters
- D) None of the above
12. Steeper transition band is present in _____ filter.
- A) Butterworth filters
- B) Chebyshev Filters
- C) Elliptic filters
- D) Kalman filter
13. Task of obtaining a digital filter from the analog IIR filter involves developing a
- A) Mapping from s-plane to z-plane
- B) Mapping from z-plane to s-plane
- C) Mapping from s-plane to s-plane
- D) Mapping from z-plane to z-plane
14. The bilinear transformation is given by
- A) $s = \frac{2}{T} \left\{ \frac{1}{1+z^{-1}} \right\}$
- B) $s = \frac{2}{T} \left\{ \frac{1+z^{-1}}{1+z^{-2}} \right\}$
- C) $s = \frac{2}{T} \left\{ \frac{1-z^{-1}}{1+z^{-1}} \right\}$
- D) $s = \frac{T}{2} \left\{ \frac{1+z^{-1}}{1+z^{-2}} \right\}$
15. Bilinear transformation method
- A) Transforms the entire left half of s-plane into the unit circle in the z-plane
- B) Transforms a strip on the left half of s-plane between $-\pi$ to π
- C) Transforms a strip on the right half of s-plane between $-\pi$ to π
- D) Transforms the entire right half of s-plane into the unit circle in the z-plane
16. Since the length of the impulse response is infinite in IIR filter, the frequency response of the filter is
- A) Discrete
- B) Periodic
- C) Continuous
- D) Non periodic
17. Which of the following windows has a time domain sequence $h(n) = 0.54 - 0.46 \cos 2\pi n/(M-1)$?
- A) Bartlett window
- B) Blackman window
- C) Hamming window
- D) Hanning window
18. Which of the following gives the equation for group delay in a FIR filter?
- A) $d\theta(\omega)/d\omega$
- B) $\theta(\omega)$
- C) $-d\theta(\omega)/d\omega$
- D) $-\theta(\omega)$
19. The format is commonly used in Modern DSP processors
- A) Floating point
- B) Fixed point
- C) IEEE floating point
- D) Variable point
20. The basic building blocks essential to carry out DSP computations will not include this
- A) Multiplier
- B) ALU
- C) Shifter
- D) USB

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Compute 8-point DFT of the sequence $x(n) = (1, 2, 3, 4, 4, 3, 2, 1)$ using DIF-FFT algorithm.	8	L*3	2	1
b)	Using relevant properties, compute N-point DFT of				
i)	$x_c(n) = x(n) \cos \frac{2\pi k_0 n}{N}$	4	L3	1	1
ii)	$x_s(n) = x(n) \sin \frac{2\pi k_0 n}{N}$				
c)	Consider the sequence $x_1(n) = \cos \frac{2\pi n}{N}$ and $x_2(n) = \sin \frac{2\pi n}{N}$; $0 \leq n \leq N-1$. Find N point circular correlation of $x_1(n)$ and $x_2(n)$.	4	L3	1	1
2. a)	For a given sequence $x(n) = (0, 1, 2, 3, 4, 5, 6, 7)$ find DFT using DIT-FFT algorithm.	8	L3	2	1
b)	$X(k)$ is a 6-point DFT of a length-6 real sequence $x(n)$ given by $x(n) = (1, -1, 2, 3, 0, 0)$. Without computing IDFT, determine the length-6 sequence $y(n)$ whose 6-point DFT is given by $Y(k) = W_3^{2k} X(k)$.	4	L3	1	1
c)	A 14-point DFT of length 14 real sequence $x(n)$ is $X(k)$. The first 8 samples of DFT is given by $X(0) = 12, X(1) = -1 + j3, X(2) = 3 + j4, X(3) = 1 - j5, X(4) = -2 + j2, X(5) = 6 + j3, X(6) = -2 - j3, X(7) = 10$. Find the remaining samples of $X(k)$. Also, evaluate the following. (i) $\sum_{n=0}^{13} x(n)$ (ii) $\sum_{n=0}^{13} x(n) ^2$	4	L3	1	1
3. a)	Compute IDFT of the DFT samples $X(k) = (4, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.414, 0, 1 + j2.414)$ using DITFFT algorithm.	8	L3	2	1
b)	$x(n)$ is a length-4 sequence with DFT $X(k) = (0, 1 + j, 1, 1 - j)$. Using relevant DFT properties, find DFT of				
i)	$x_2(n) = \cos\left(\frac{\pi}{2}n\right)x(n)$	4	L3	1	1
ii)	$x_3(n) = x((n-1))_4$				
c)	Using Stockham's method, compute the circular convolution of $x_1(n) = (2, 1, 2, 1)$ and $x_2(n) = (1, 2, 3, 4)$.	4	L3	1	1
Unit – II					
4. a)	For a given system function, obtain the difference equation representation and then illustrate the direct form I and direct form-II realization. Given $H(z) = \frac{3+3.6z^{-1}+0.6z^{-2}}{1+0.1z^{-1}-0.2z^{-2}}$	8	L3	4	1
b)	Find the order, cut-off frequency and poles of a Butterworth LPF and plot the poles. The specifications are $\Omega_p = 20 \frac{\text{rad}}{\text{sec}}, \Omega_s = 30 \frac{\text{rad}}{\text{sec}}, K_p = -2 \text{ dB}, K_s = -10 \text{ dB}$.	8	L3	3	1
5. a)	An LTI system has an impulse response $h(n) = 2\delta(n) - \delta(n-1)$. Determine the output of the system for the input sequence $x(n) = \delta(n) + 3\delta(n-1) + 2\delta(n-2) - \delta(n-3) + \delta(n-4)$ using (i) Linear convolution (ii) Circular convolution (iii) Linear by circular convolution. Compare the results obtained in (i) and (ii) and show the presence of aliasing.	8	L3	3	1

- b) Design a Butterworth low passband filter with passband attenuation as -1 dB and PB frequency as 4 rad/sec. SB attenuation is -20 dB with Stop band frequency as 8 rad/sec.

8 L3 3 1

6. a) Design a digital Butterworth filter with passband edge frequency as 500 Hz, passband attenuation as -3.01 dB. Stop band edge frequency as 750 Hz, stopband attenuation as -15 dB. The sampling rate is 2 kHz. Use Bilinear transformation method.

8 L3 4 1

- b) Find the output $y(n)$ of a filter with impulse response $h(n) = (1, 1, 1)$ and input signal $x(n) = (3, -1, 0, 1, 3, 2, 0, 1, 2, 1)$ using overlap-save method choosing a block length of 5.

8 L3 3 1

Unit – III

7. a) The desired frequency response of an LPF is

$$H_d(e^{j\omega}) = e^{-j3\omega}; -\frac{3\pi}{4} \leq \omega \leq \frac{3\pi}{4}$$

$$= 0; \frac{3\pi}{4} < |\omega| \leq \pi$$

Design it using Hamming window ($M=7$) and obtain the system function.

8 L3 5 1

- b) Describe Braun Multiplier for unsigned number multiplication. Highlight the issues observed in multiplication process.

4 L2 5 1

- c) An FIR filter is given by the difference equation as $y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$. Realize this using lattice structure.

4 L3 5 1

8. a) Design a low pass filter using rectangular window sequence of length $M=11$ for the frequency response

$$H_d(e^{j\omega}) = e^{-j5\omega}; |\omega| \leq \frac{\pi}{2}$$

$$= 0; \frac{\pi}{2} \leq |\omega| \leq \pi$$

Obtain the impulse response and transfer function.

8 L3 5 1

- b) With a neat connection diagram explain Barrel shifter.

4 L2 5 1

- c) Obtain the direct form realization of linear phase FIR filter for the system function

i) $H(z) = \frac{1}{2} + \frac{1}{7}z^{-1} + \frac{1}{6}z^{-2} + \frac{1}{6}z^{-3} + \frac{1}{7}z^{-4} + \frac{1}{2}z^{-5}$.

ii) $H(z) = 1 + \frac{1}{3}z^{-1} + \frac{1}{5}z^{-2} + \frac{1}{4}z^{-3} + \frac{1}{8}z^{-4} + \frac{1}{9}z^{-5} + \frac{1}{2}z^{-6}$.

4 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
